

THE PIGEON HOLE PRINCIPLE

1. WHAT IS THE PIGEON HOLE PRINCIPLE?

1.1. The Pigeon-Hole Principle. If n objects are placed in r boxes, and $r < n$, then at least one of the boxes contains more than one object.

Example 1.1.1. If there are 5 pigeons but only 4 pigeon holes, then two pigeons must be in the same hole.

2. HOW CAN WE USE IT?

In each of the following proofs, keep the following questions in mind: **How are we applying the pigeon hole principle here? What are the pigeons and what are the pigeon holes?** Also, come up with examples illustrating each theorem.

Theorem 2.0.1. *If you pick five cards from a standard deck of 52 cards, then at least two will be of the same suit.*

Proof. Use the pigeon hole principle! □

Theorem 2.0.2. *In a city of 3 million people, there are two people with the same number of hairs on their head (assuming the maximum number of hairs on a human head is 500000.)*

Proof. Use the pigeon hole principle! □

Theorem 2.0.3. *If you pick any distinct 5 numbers from the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$, then there must be two which add up to 9.*

Proof. First, pair up the numbers in the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$ so that the sum of each pair adds up to 9. How many pairs are there? Then, use the pigeon hole principle! Bonus, generalize this result appropriately for the set $\{1, 2, 3, \dots, 2n\}$. □

Theorem 2.0.4. *Suppose that A is a set contained in $\{1, 2, \dots, 2n\}$ of size $n + 1$. Then there are always two numbers in A such that one divides the other.*

Before we prove the theorem, let's do an example with $n = 4$. So, let's take a set A contained in the set $\{1, 2, 3, 4, 5, 6, 7, 8\}$. For example, suppose A is the set $\{1, 2, 3, 5, 8\}$. We see that 2 divides 8, so the theorem holds! Now, let's prove it for any n .

Proof. Step 1: Write every number a in the form $a = 2^k m$, where m is an odd number between 1 and $2n - 1$.

Step 2: Let $M = \{1, 3, \dots, 2n - 1\}$ be the set of these odd numbers, and note that M contains exactly n numbers.

Step 3: Since there are $n + 1$ numbers in A and only n numbers in M , there must be a_1, a_2 in A such that $a_1 = 2^k m$ and $a_2 = 2^j m$. If $a_1 < a_2$, then a_1 divides a_2 . Otherwise, a_2 divides a_1 . □

Remark 2.0.5. This was one of Paul Erdos' favorite "initiation to mathematics" problems – he posed it to young Lajos Posa during a dinner, who figured it out by the end of the night.